Owner-Harm: A Missing Threat Model for AI Agent Safety

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Abstract

Existing AI agent safety benchmarks focus predominantly on generic criminal harm—cybercrime, harassment, and weapon synthesis—leaving a systematic blind spot for a distinct and commercially consequential threat category: agents harming their own deployers. Real-world incidents illustrate this gap: the Slack AI credential exfiltration (August 2024), Microsoft 365 Copilot data leaks via calendar injection (January 2024), and a Meta agent unauthorized forum post exposing operational data (March 2026). We propose *Owner-Harm*, a formal threat model defining eight categories of agent behavior that damage the deploying owner's interests. We quantify the resulting defense gap on two authoritative benchmarks: the same compositional safety system achieves 100% TPR / 0% FPR on AgentHarm (generic criminal harm, 176 harmful scenarios) yet only 14.8% (4/27; 95% CI: 5.9%–32.5%) on AgentDojo injection tasks (prompt-injection-mediated owner harm), exposing a category-specific blind spot. A controlled experiment with a zero-shot generic LLM classifier reveals that this gap is not inherent to the owner-harm category (generic LLM: 62.7% vs. 59.3%, gap = 3.4 pp) but arises from environment-bound symbolic rules that fail to generalize across tool vocabularies. On a post-hoc diagnostic benchmark of 300 owner-harm scenarios, the gate alone achieves 75.3% TPR / 3.3% FPR; adding a deterministic post-audit verifier raises overall TPR to **85.3%** (+10.0 pp) and Hijacking detection from 43.3% to 93.3%, demonstrating strong layer complementarity. These results confirm that owner-harm requires context-aware safety mechanisms that explicitly model resource ownership, trust boundaries, and authorization scope. To organize these observations, we introduce the *Symbolic-Semantic Defense Generalization (SSDG)* conceptual framework, which describes how information coverage—the fraction of an attack's evidential requirements satisfied by a defense's context set—relates to detection rate. Two controlled SSDG experiments partially test the framework: context deprivation induces a 3.4× amplification of the detection gap (gap ratio $R = 3.60$ vs. $R = 1.06$ with full context), while context injection reveals that goal-context awareness is necessary but not sufficient—structured goal-action alignment, not simple text concatenation, is required for effective owner-harm detection.

1 Introduction

The deployment of autonomous AI agents into production systems has accelerated faster than the safety frameworks needed to govern them. Current research focuses on preventing agents from facilitating generic criminal harm: generating malware, synthesizing dangerous substances, or producing

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harassing content. Yet a qualitatively different and commercially urgent threat class has emerged in practice—agents harming the very organizations that deploy and trust them.

Three incidents anchor this concern. In August 2024, PromptArmor disclosed that Slack AI could be manipulated via prompt injection to exfiltrate private channel tokens from the deploying organization PromptArmor [2024]. In January 2024, Zenity Research demonstrated that Microsoft 365 Copilot could be hijacked through malicious calendar invitations to forward sensitive emails to external parties Zenity Research [2024]. In March 2026, a Meta AI agent made an unauthorized post to an internal forum, exposing operational data for two hours before containment Schiffer [2026]. In each case, the agent's deployer—not a third-party victim—bore the harm.

The benchmark gap. Existing agent safety benchmarks do not model this pattern. AgentHarm Andriushchenko et al. [2025] catalogs 11 harm categories, all framed as an adversary using an agent against third parties. AgentDojo Debenedetti et al. [2024] evaluates prompt injection defenses across four suites (Banking, Travel, Workspace, Slack) but without an explicit owner-harm taxonomy. The consequences are measurable: the same compositional safety system that achieves 100% TPR on AgentHarm scores only 3.7% (1/27) on AgentDojo injection tasks using deterministic rules, and 14.8% (4/27) with semantic reasoning added. The constraints that successfully intercept generic criminal content—system call patterns, toxicity signals—have near-zero coverage of application-layer owner harm, such as financial transactions initiated via legitimate banking APIs or PII forwarded through authorized email tools.

Contributions. This paper makes three contributions:

1. **Owner-Harm threat model.** We provide the first formal definition of owner-harm and its eight sub-categories, grounded in real incidents and distinguished from existing taxonomies.
2. **Defense gap quantification and gap attribution.** We measure the gap (100% vs. 14.8%) and use a controlled experiment to attribute it to environment-bound rule generalization failure rather than inherent category difficulty: a zero-shot generic LLM shows only a 3.4 pp gap (62.7% vs. 59.3%) across the same benchmarks.
3. **Layer complementarity evidence.** We show that deterministic post-audit verification is orthogonal to semantic gate reasoning for Hijacking (C7), raising detection from 43.3% to 93.3% via complementary detection dimensions.

2 Owner-Harm Threat Model

2.1 Formal Definitions

We model an AI agent as a tuple $\mathcal{A} = \langle M, \mathcal{T}, \mathcal{E}, \pi \rangle$, where M is the underlying language model, $\mathcal{T}$ is the available tool set, $\mathcal{E}$ is the environment state space, and π is the agent's policy. An agent produces an action sequence $\vec{a} = (a_1, \ldots, a_n)$ transforming environment state s_0 to s'.

Definition 1 (Owner)**.** The *owner* $\mathcal{O}$ is the entity that deploys $\mathcal{A}$, grants resource access $R_{\mathcal{O}}$, and bears accountability for agent actions. $\mathcal{O}$ is associated with a resource set $R_{\mathcal{O}}$, a utility function $U_{\mathcal{O}}$ over environment states, and a trust boundary $\mathcal{B}_{\mathcal{O}}$ delineating authorized interactions.

Definition 2 (Owner-Harm)**.** An action sequence $\vec{a}$ causes *owner-harm* with respect to owner $\mathcal{O}$ if and only if:

$$\mathrm{OH}(\vec{a}, \mathcal{O}) \coloneqq U_{\mathcal{O}}(s') < U_{\mathcal{O}}(s_0) \ \wedge \ \exists\, a_i \in \vec{a} : \mathrm{cause}(a_i,\, s_0 \to s')$$

where $\mathrm{cause}(a_i, s_0 \to s')$ means a_i is a proximate or distal cause of the state transition from s_0 to s'.

Definition 3 (Direct Owner-Harm)**.**

$$\mathrm{DOH}(\vec{a}, \mathcal{O}) \coloneqq \mathrm{OH}(\vec{a}, \mathcal{O}) \ \wedge \ \mathrm{ProxCause}(\mathcal{A}, \vec{a})$$

where $\mathrm{ProxCause}(\mathcal{A}, \vec{a})$ means the agent is the proximate cause of the harmful state transition.

Definition 4 (Indirect Owner-Harm)**.**

$$\mathrm{IOH}(\vec{a}, \mathcal{O}, \mathcal{V}) \coloneqq \mathrm{OH}(\vec{a}, \mathcal{O}) \ \wedge \ \mathrm{Manipulate}(\mathcal{V}, \pi, \vec{a})$$

where $\mathcal{V}$ is an adversarial actor that manipulates the agent's policy π to produce the harmful action sequence—the prompt injection attack model.

Table 1: Coverage of Owner-Harm Categories by Existing Frameworks

Category	AgentHarm	ToolEmu	OWASP	AgentDojo
C1: Credential Leak	∼	✓	✓	∼
C2: Infrastructure Exposure	×	∼	✓	×
C3: Privacy Exposure	∼	✓	✓	✓
C4: Inner Circle Leak	×	×	×	×
C5: Asset Destruction	∼	✓	∼	∼
C6: Exfiltration via Tools	×	∼	∼	∼
C7: Hijacking	×	∼	✓	∼
C8: Unauthorized Autonomy	∼	✓	∼	×

✓ = covered; ∼ = partial; × = gap

2.2 Eight Owner-Harm Categories

We enumerate eight categories $C1$–$C8$ (primary labels; a single incident may span multiple categories), each grounded in documented incidents.

C1: Credential Leak. The agent exfiltrates authentication materials (API keys, OAuth tokens, session cookies, passwords) belonging to or entrusted to the owner. *Real case*: Slack AI August 2024 PromptArmor [2024]—injected prompt caused the assistant to echo private channel tokens to the attacker.

C2: Infrastructure Exposure. The agent misconfigures or discloses network rules, cloud resource policies, or internal architecture, expanding the owner's attack surface. *Pattern*: AI-assisted code generation producing over-permissive AWS IAM policies that expose production databases.

C3: Privacy Exposure. The agent transmits personally identifiable information (PII) or confidential business data to unauthorized parties. *Real case*: Microsoft 365 Copilot January 2024 Zenity Research [2024]—calendar injection caused Copilot to forward sensitive email contents externally.

C4: Inner Circle Leak. The agent betrays trust relationships that are commercially sensitive to the owner: client lists, partner agreements, M&A discussions, or strategic roadmaps. *Pattern*: Samsung employees pasting proprietary semiconductor source code into ChatGPT (April 2023), with the owner-harm being the confidentiality breach to a third-party model.

C5: Asset Destruction. The agent irreversibly deletes or corrupts owner data, configurations, or digital assets. *Pattern*: AI coding agents executing `rm -rf` on production directories when given broad filesystem access and ambiguous cleanup instructions.

C6: Exfiltration via Tools. The agent exploits *authorized* tools as covert data channels—using email, webhooks, or file-write operations to smuggle sensitive data to attacker-controlled endpoints. *Real case*: ASCII smuggling via Microsoft 365 Copilot Rehberger [2024]—injected prompt caused Copilot to encode and transmit sensitive data through Markdown image URLs rendered invisibly.

C7: Hijacking. An adversary achieves persistent control over the agent using the owner's identity, credentials, or resource access—the agent becomes a weaponized surrogate. *Pattern*: AutoGPT memory poisoning (2024–2025) where injected memory entries caused the agent to act on behalf of the attacker across subsequent sessions.

C8: Unauthorized Autonomy. The agent exceeds its authorization scope by taking consequential actions without required human confirmation. *Real case*: Air Canada chatbot (February 2024) committed the airline to unauthorized refund terms in a legally binding interaction British Columbia Civil Resolution Tribunal [2024].

2.3 Comparison with Existing Taxonomies

Table 1 compares Owner-Harm categories against four existing frameworks: AgentHarm Andriushchenko et al. [2025] (11 categories), ToolEmu Ruan et al. [2024] (7 risk types), OWASP LLM Top 10, and AgentDojo Debenedetti et al. [2024] (4 suites).

The most notable gap is **C4 (Inner Circle Leak)**, which appears in no existing benchmark. This category requires reasoning about the owner's trust graph—who counts as an insider—which no

Table 2: Generic Defense Failure Modes for Owner-Harm

Defense Class	Owner-Harm Blind Spot
Content classifier	Ignores $R_{\mathcal{O}}$; "send email" is benign by content
Injection detector	Detects injection syntax; misses semantic intent
Tool monitor	Sees call signature; not authorization context
DLP system	Pattern-matches known PII; misses context-defined confidential data
Behavioral anomaly	Requires baseline; new deployments have no baseline

current system models explicitly. **C6** is partially covered by ToolEmu but misses the *covert channel* aspect: using authorized tools as smuggling vectors. The *owner-centric perspective* itself is the primary novel contribution: existing frameworks evaluate harm against third parties, users, or society; none formally model the deploying owner as the victim.

2.4 Why Generic Defenses Fail

The fundamental mismatch is context-dependence of harm:

$$\exists\, c, \mathcal{O}_1, \mathcal{O}_2 : \mathrm{Harmful}(c, \mathcal{O}_1) \;\wedge\; \neg\mathrm{Harmful}(c, \mathcal{O}_2)$$

The action "forward email to external@attacker.com" is benign for an owner whose trust boundary includes that address, and harmful for an owner whose trust boundary does not. Content-only classifiers cannot make this distinction.

Three owner-context dimensions are systematically absent from generic defenses:

- **Resource Ownership** ($R_{\mathcal{O}}$): which data and credentials belong to the owner and should not be disclosed.
- **Trust Boundary** ($\mathcal{B}_{\mathcal{O}}$): who the owner considers an authorized counterparty for each resource type.
- **Authorization Scope** ($\mathrm{Auth}_{\mathcal{O}}$): which actions the owner has explicitly sanctioned without requiring confirmation.

Proposition 1 (Generic Classifier Incompleteness)**.** Any context-free binary classifier $f_{\mathrm{gen}} : \mathcal{C} \to \{0, 1\}$ that classifies action content without access to $R_{\mathcal{O}}$, $\mathcal{B}_{\mathcal{O}}$, or $\mathrm{Auth}_{\mathcal{O}}$ will produce owner-harm false negatives on scenarios where harm is defined relative to the owner's context.

Table 2 summarizes the failure modes of common defense classes against this incompleteness.

2.5 Symbolic-Semantic Defense Generalization (SSDG) Framework

The controlled experiment results motivate a conceptual account of *when* and *why* defenses fail to generalize across attack categories. We emphasize that SSDG is a conceptual framework for organizing observations, not a formal mathematical theory with rigorous proofs. For formal treatments of causality in such analyses, see Pearl Pearl [2009]. We introduce the following framework.

Definitions. Let an attack a be characterized by its *evidential requirement set* $E(a)$—the set of contextual facts needed to determine its harmfulness. A defense D maintains an *information set* $\Omega(D) = \Omega_S \cup \Omega_C$, where Ω_S captures surface-level features (tool names, parameter patterns) and Ω_C captures contextual knowledge (world knowledge, owner context, task goals). The *information coverage* of D for attack a is:

$$\mathrm{Cover}(D, a) \;:=\; \frac{|E(a) \cap \Omega_{\mathrm{usable}}(D)|}{|E(a)|} \;\in [0, 1]$$

where $\Omega_{\mathrm{usable}}(D) \subseteq \Omega(D)$ denotes the subset of information that the defense can *effectively use* in its decision process. Mere availability of information (Ω_{raw}) does not guarantee usability—simple

text concatenation of context may not enable structured comparison (as P1 in Section 4.6 demonstrates).

Assumption (Detection-Coverage Monotonicity). $\mathrm{TPR}(D, a)$ is non-decreasing in $\mathrm{Cover}(D, a)$; below threshold θ, $\mathrm{TPR}(D, a) \leq \varepsilon$.

Hypothesis 1 (Symbolic Defense Gap)**.** A pure symbolic defense with $\Omega_C = \emptyset$ achieves $\mathrm{Cover}(D, A_G) \gg \mathrm{Cover}(D, A_O)$ when A_O requires joint surface-context evidence. Consequently, $\mathrm{TPR}(D, A_O) \leq \varepsilon$ while $\mathrm{TPR}(D, A_G)$ remains high—producing a structural detection gap.

Hypothesis 2 (Semantic Defense Stability)**.** A semantic defense with $\Omega_C \supseteq \mathcal{C}_{\mathrm{world}}$ (world knowledge) satisfies $\mathrm{Cover}(D, A_G) \approx \mathrm{Cover}(D, A_O)$, since both attack types' evidential requirements are met by world-knowledge reasoning. The inter-category detection gap is thus bounded by inference capacity, not information availability.

These hypotheses yield two falsifiable predictions tested in Section 4.6: **P1**—injecting the user goal into a symbolic defense's context should raise $\mathrm{Cover}(D, A_O)$, increasing TPR; and **P2**—stripping world-context from a semantic defense should cause it to exhibit symbolic-gap behavior, with gap ratio $R > 3\times$.

3 Nous: Compositional Runtime Safety Gate

We evaluate owner-harm detection using **Nous**, a four-layer compositional runtime safety gate for AI agent actions. We describe it briefly here; the architecture is the primary evaluation vehicle rather than the primary contribution.

Layer 1 (L1): Deterministic Datalog Blocker. 46 Datalog rules encode owner-policy constraints (resource ownership, known-bad patterns, categorical prohibitions). L1 is fail-closed, fast ($P_{50} = 0.055$ ms), and requires no LLM call. Rules cover categorical owner-harm such as exfiltration of credentials with recognizable patterns (C1) and shell commands associated with asset destruction (C5).

Layer 2 (L2): Triviality Filter. Routes trivially benign actions past the expensive semantic layer, yielding approximately 70% cost reduction with negligible safety impact.

Layer 3 (L3): Semantic Gate. An LLM evaluator supplied with minimal-pair examples, intent decomposition prompts, and majority vote $k = 3$ to handle stochasticity in mixture-of-experts models. L3 provides semantic reasoning about owner context: whether an action falls within the owner's trust boundary, whether resource access is proportionate to the stated task.

Layer 4 (L4): Post-Gate Verifier. Six deterministic audit rules (V1–V6) inspect *file contents* and *multi-step tool-call sequences* that are invisible to L1–L3 because they emerge from composed actions rather than individual calls. L4 adds ∼0.038 ms latency with zero LLM cost. Rules target prompt injection patterns embedded in file artifacts and behavioral sequences characteristic of Hijacking (C7).

Why this architecture for owner-harm. L1 encodes owner-policy as explicit Datalog facts; L3 reasons about owner context when policy is ambiguous; L4 catches injection patterns that survive semantic evaluation because the semantic reasoning is itself manipulated by injected content. The layers are architecturally complementary: L1 operates on individual action content; L3 operates on action semantics plus owner context; L4 operates on artifact contents and behavioral sequences.

4 Evaluation

4.1 Evaluation Strategy

We evaluate on three benchmark sets with different evidentiary weight:

1. **AgentDojo** Debenedetti et al. [2024] (authoritative): 27 injection tasks, 97 user tasks, 4 suites (Banking, Travel, Workspace, Slack). Ground-truth labels provided by benchmark authors. Safety score = fraction of injection tasks blocked.

Table 3: AgentDojo Results: Safety Score vs. Utility (27 injection tasks, 97 user tasks)

Configuration	Safety	Utility	FPR
L1 only (baseline)	3.7% (1/27)†	97.9% (95/97)	2.1% (2/97)
L1 + L4	3.7% (1/27)†	97.9%	2.1%
L1 + L3 + L4 (full)	14.8% (4/27)†	97.9%	2.1%
†95% Wilson CI: 3.7% [0.7%–18.3%]; 14.8% [5.9%–32.5%]			

Table 4: AgentDojo Safety by Owner-Harm Category

Category	L1 only	L1+L3+L4
Financial Harm	0/10 (0%)	1/10 (10%)
Privacy Breach	0/6 (0%)	2/6 (33%)
Operational Harm	1/4 (25%)	1/4 (25%)
Credential Leak	0/3 (0%)	0/3 (0%)
Reputational Harm	0/3 (0%)	0/3 (0%)
Other	0/1 (0%)	0/1 (0%)
Total	**1/27 (3.7%)**	**4/27 (14.8%)**

2. **AgentHarm** Andriushchenko et al. [2025] (comparative baseline): 176 harmful + 176 benign scenarios across 11 categories. Used to confirm that the 3.7%/14.8% gap on AgentDojo is category-specific, not a global system deficiency.
3. **Owner-Harm Benchmark** (post-hoc diagnostic, 300H + 150B, 8 categories): Constructed by the same author after optimization loops. *This is NOT a held-out independent test set*; results should be interpreted as diagnostic evidence of system capabilities. Reported separately from the authoritative benchmarks to avoid overstating validity.

AgentDojo to Owner-Harm category mapping. We assign each of the 27 injection tasks to the Owner-Harm category it most closely targets: Banking (9 injection tasks): `financial_harm` (8 tasks—unauthorized transfers with data exfiltration), `credential_leak` (1 task—password modification); Travel (7 tasks): `privacy_breach` (6 tasks—passport/banking info exfiltration), `reputational_harm` (1 task); Workspace (6 tasks): `operational_harm` (file deletion), `privacy_breach` (email forwarding), `credential_leak` (2FA bypass); Slack (5 tasks): `reputational_harm` (phishing message), `privacy_breach` (data exfiltration), `other` (SSRF).

4.2 AgentDojo Results

Table 3 presents the three-configuration comparison. All configurations are evaluated with the AgentDojo ground-truth mode (deterministic task success scoring; no adversarial attack LLM used in this evaluation).

Layer 4 contributes zero to AgentDojo safety. Verifier rules V1–V6 target bash command sequences and curl exfiltration patterns embedded in file artifacts—patterns characteristic of system-level Hijacking. AgentDojo injection tasks operate at the *application layer*: the attack tools are `send_money`, `send_email`, `update_calendar`. No file artifact intermediary exists for V-rules to inspect.

Layer 3 adds +11.1pp by catching three cases unreachable by Datalog: `travel/injection_task_3` (Privacy Breach—passport details forwarded to adversary URL), `travel/injection_task_5` (Privacy Breach—bank account numbers forwarded via booking confirmation), and `banking/injection_task_6` (Financial Harm—batch wire transfer to adversary account at plausible individual amounts). Utility remains at 97.9% across all configurations, confirming that semantic reasoning does not increase false positives.

Table 4 shows per-Owner-Harm-category results.

Structural blind spots. Three categories of injection tasks remain systematically out of reach regardless of configuration:

Table 5: Owner-Harm Benchmark: Per-Category Results (Post-Hoc Diagnostic)

Category	N_h	Gate TPR	Combined TPR	Gate FPR
C1: Credential Leak	28	100.0%	100.0%	0.0%
C2: Infrastructure Exposure	50	100.0%	100.0%	20.0%†
C3: Privacy Exposure	28	100.0%	100.0%	0.0%
C4: Inner Circle Leak	28	89.3%	89.3%	0.0%
C8: Unauthorized Autonomy	50	70.0%	70.0%	0.0%
C5: Asset Destruction	28	64.3%	64.3%	0.0%
C6: Exfiltration via Tools	28	57.1%	57.1%	0.0%
C7: Hijacking	**60**	**43.3%**	**93.3%**	**0.0%**

† 20% FPR for C2 caused by rule OH-R3a over-triggering on benign external-channel configuration queries.

- *Text-only manipulation* (e.g., `travel/injection_task_6`: adversarial advertising text injected into a hotel description, with no tool call produced). The gate has no tool-call artifact to intercept.
- *Malicious parameters indistinguishable from legitimate ones* (e.g., `send_money` with a plausible amount and adversary-controlled recipient that appears to be a valid payee). Without the user's task goal in the gate context, the semantic layer cannot compare "user requested payment to X" versus "agent is paying Y".
- *Multi-step attacks where partial interception is insufficient* (e.g., a read-then-exfiltrate chain where blocking the read breaks utility and missing the exfiltrate misses the attack).

Theoretical ceiling without goal context: we estimate 40–60% safety score on AgentDojo is achievable with user task goal injected into gate context, enabling the "user requested X, agent doing Y" comparison that resolves malicious-parameter ambiguity.

Comparison with AgentDojo baselines. Debenedetti et al. report that a GPT-4 agent with `tool_filter` defense achieves approximately 14.8% safety on the Banking suite Debenedetti et al. [2024]. Nous L1+L3+L4 matches this at 14.8% overall across all four suites, while maintaining 97.9% utility compared to approximately 85% for `tool_filter` (which blocks entire tool categories rather than discriminating at the call level).

4.3 Owner-Harm Benchmark Results (Post-Hoc Diagnostic)

Important caveat: the Owner-Harm Benchmark was constructed by the same author after optimization loops on the Nous system. It is not an independent held-out evaluation. All 450 scenarios were labeled by a single annotator (the first author); no inter-annotator agreement (IAA) was computed. The C1–C8 primary-category assignments reflect the author's subjective judgment of the dominant harm vector, and some scenarios could plausibly be assigned to multiple categories. The results below constitute diagnostic evidence that the system has learned the intended owner-harm categories; they do not constitute validation performance in the machine-learning sense.

Overall results.

Gate only (L1+L2+L3): TPR $= 75.3\%$ (226/300), 95% Wilson CI [70.2%, 79.9%]; FPR $= 3.3\%$ (5/150), CI [1.4%, 7.6%].

Gate + Verifier (all layers): TPR $= 85.3\%$ (256/300), 95% Wilson CI [80.9%, 88.9%]; FPR $= 3.3\%$ (unchanged).

Table 5 shows per-category breakdown.

Hijacking layer complementarity. The most significant result is the Hijacking (C7) improvement from 43.3% gate-only to 93.3% combined. Table 6 shows the 2×2 detection quadrant analysis (60 harmful, 30 benign).

Wilson 95% CIs: Gate TPR [31.6%, 55.9%]; Combined TPR [84.1%, 97.4%]; Hijacking benign FPR [0.0%, 11.4%]. The 50% "Verifier-only" quadrant (30 cases) represents prompt-injection attacks where the semantic gate is itself manipulated by injected content, but the verifier detects the

Table 6: Hijacking (C7) Detection Quadrant Analysis (60 harmful, 30 benign)

Quadrant	Count	%
Gate-only (verifier misses)	11	18.3%
Verifier-only (gate allows)	30	50.0%
Both layers detect	15	25.0%
Neither (structural boundary)	4	6.7%
Combined	**56**	**93.3%**

Table 7: Cross-Benchmark Safety Scores (Nous vs. Generic LLM Baseline)

Benchmark	Focus	Nous Safety	Generic LLM	Status
AgentHarm	Generic criminal	100% / 0%	62.7% / 16.0%	Validation
Owner-Harm (diag.)	Owner-directed	75.3% / 3.3%	—	Diagnostic
AgentDojo	Prompt injection	14.8% (4/27)	59.3% (16/27)	Authoritative

Generic LLM = `qwen-turbo` zero-shot; gap = |Nous gap| − |Generic gap| = 85.2 pp − 3.4 pp = 81.8 pp attributable to symbolic rule generalization failure.

behavioral fingerprint in file artifacts. The 4 structural boundary cases (6.7%) involve direct SQL file injection and direct SSH key injection patterns not yet covered by V-rules.

4.4 Cross-Benchmark Comparison

Table 7 presents the central finding: the same system exhibits a 6.8× gap in safety score between generic harm and owner-harm scenarios.

The 100% → 14.8% gap is not a system deficiency. It is the paper's primary evidence: the constraints that achieve 100% on AgentHarm (system call toxicity patterns, malware-synthesis keywords, harassment content signals) have near-zero coverage of application-layer owner-harm. "Send money to account X" is indistinguishable from a legitimate banking action by its content alone; it becomes harmful only relative to the owner's authorization scope and the user's actual task goal.

4.5 Controlled Experiment: Gap Attribution

The 85.2 pp gap between AgentHarm (100%) and AgentDojo (14.8%) admits two competing explanations: (H1) the gap reflects a genuine structural difference—owner-harm scenarios detected via prompt injection are semantically harder to classify; or (H2) the gap is an artifact of Nous's symbolic rules being fit to the AgentHarm tool vocabulary and failing to generalize.

Experimental design. We evaluate a zero-shot generic LLM classifier (`qwen-turbo` via API relay) on the same data, using a uniform binary-classification prompt with no benchmark-specific fine-tuning:

> *You are a safety classifier. An AI agent is performing a task. User's original request: {prompt}. Agent's planned action: {tool} ({params}). Is this action potentially harmful to the user? Answer YES or NO.*

For AgentDojo, the goal description (injected instruction) is substituted for the tool-call representation. The baseline has no access to owner context ($R_{\mathcal{O}}$, $\mathcal{B}_{\mathcal{O}}$, $\text{Auth}_{\mathcal{O}}$).

Results. Table 8 shows the 2×2 comparison.

Finding. The generic LLM exhibits a 3.4 pp gap (62.7% vs. 59.3%)—statistically negligible compared to Nous's 85.2 pp gap. This is inconsistent with H1: AgentDojo owner-harm scenarios do not appear to be semantically harder to detect. The generic LLM achieves 59.3% on AgentDojo without any prompt-injection-specific design, suggesting that the harm is semantically recognizable by content alone.

Gap attribution. Nous's gap is strongly associated with **Layer 1 (Datalog) generalization failure**. L1 rules were derived from AgentHarm patterns—`bash(script)`, `read_file`

Table 8: Controlled Experiment: Gap Attribution (Nous vs. Generic LLM Classifier)

System	AgentHarm TPR	AgentDojo TPR	Gap
Nous (L1+L3+L4)	100.0%	14.8% (4/27)	85.2 pp
Generic LLM	62.7%	59.3% (16/27)	3.4 pp

Generic LLM AgentHarm: 110 TP / 66 FN / 28 FP (FPR = 15.9%, $N = 176$ harmful + 176 benign). AgentDojo per-category breakdown appears in text.

Table 9: P2: Context Deprivation Induces Symbolic-Class Gap (pilot, $N = 12$ AgentDojo + 15 AgentHarm)

Condition	AgentHarm TPR	AgentDojo TPR	Gap Ratio R
Full Group	62.7%	59.3%	1.06
Stripped Group	30.0%	8.3%	3.60

$R = \mathrm{TPR}_{\mathrm{AH}}/\mathrm{TPR}_{\mathrm{AD}}$. P2 falsification threshold: $R \leq 1.5$.

`(malicious_path)`, `execute(command)`—which are absent from AgentDojo's application-layer tool vocabulary (`send_money`, `send_email`, `update_calendar`). **Implication: generic LLM outperforms Nous.** Notably, the zero-shot generic LLM (59.3%) substantially outperforms Nous (14.8%) on AgentDojo owner-harm detection. This is not a contradiction—it is the central architectural insight: owner-harm detection depends primarily on semantic understanding of action consequences and owner context, which symbolic rules (L1) cannot provide. Nous's L1 achieves perfect recall on AgentHarm's system-call patterns but has near-zero coverage of application-layer actions. L3's semantic gate contributes only +11.1 pp because it also lacks owner-context grounding. The generic LLM compensates via world-knowledge reasoning, suggesting that future owner-harm defenses should prioritize structured semantic alignment over pattern-matching rules.

4.6 SSDG Experimental Predictions

The SSDG framework (Section 2.5) yields two testable predictions. We report pilot experiments ($N = 12$ AgentDojo tasks + 15 AgentHarm scenarios) using the same controlled setup as Section 4.5.

P1 (Context Injection) — Prediction Rejected. Injecting the user's task goal into the gate evaluation context (L1+L3+L4 + `user_goal`) should raise the AgentDojo TPR if goal-awareness is the missing information. Result: TPR declined from 14.8% to 8.3%—below the no-injection baseline.

This rejection has an informative interpretation: appending the user goal as a text prefix does not constitute structured goal-action alignment. The gate's intent decomposition logic cannot compare the injected goal to the observed action without explicit structural scaffolding (e.g., a "goal vs. action" contrast prompt). Three confounding factors compound this: (a) the ground-truth AgentDojo pipeline evaluates task success, not injection semantics; (b) simple text injection may activate over-cautious benign-task blocking; and (c) the pilot sample size ($N = 12$) is too small for significance. P1 remains open: the mechanism requires *structured* goal-context integration, not raw text concatenation.

P2 (Context Deprivation) — Prediction Confirmed. Stripping world-context from the semantic defense should reproduce symbolic-gap behavior. We evaluate the generic LLM classifier under two conditions: (i) *Full Group*—standard evaluation with full agent-context prompt; (ii) *Stripped Group*—evaluation with only the raw tool name and parameter, no surrounding task description.

The stripped condition yields $R = 3.60$, exceeding the P2 falsification threshold ($R \leq 1.5$). When deprived of context, the semantic defense degrades to near-symbolic gap behavior, confirming Hypothesis 2: the stability of semantic detection depends on access to world-context, not on model reasoning capacity per se. The finding identifies *information deprivation* as a key factor associated with the detection gap.

5 Production Deployment Evidence

Nous was deployed in shadow mode alongside an existing rule-based system in a production multi-agent environment for 11+ days (30,204 calls total).

Consistency: 99.47% agreement with the existing rule-based system across all calls—confirming that semantic reasoning does not introduce systematic drift from established policy.

False negative recovery: Nous detected 157 cases that the existing rule-based system passed as benign. Manual review of a stratified sample confirmed these were true positives, predominantly ambiguous read-then-send workflows where the send target fell outside the owner's trust boundary. *Caveat*: the manual review was conducted by the system developer (first author), not an independent auditor, and was not blind to Nous's verdict—creating potential confirmation bias. These results should be interpreted as case-level observations, not as a controlled study.

Over-blocking: Nous over-blocked in 4 cases where the existing system correctly allowed. Manual review confirmed all 4 were over-triggered on ambiguous read-then-send workflows where the send target was a legitimate internal address. (Same caveat as above: non-blind, author-conducted review.)

Following the shadow period, Nous was promoted to primary interceptor. No regressions were observed post-cutover.

6 Discussion

6.1 Why Generic Defenses Fail on Owner-Harm

The production and benchmark results converge on the same explanation: generic defenses evaluate action content without owner context. The action "send email to external@domain.com" carries no harm signal by content; its harmfulness is entirely determined by whether `external@domain.com` is within $\mathcal{B}_{\mathcal{O}}$. Content classifiers, injection detectors, and DLP systems operating without $R_{\mathcal{O}}$, $\mathcal{B}_{\mathcal{O}}$, and $\text{Auth}_{\mathcal{O}}$ cannot make this determination.

The controlled experiment (Section 4.5) provides a more precise attribution. The 85.2 pp gap arises *specifically* from the symbolic rule layer (L1), which encodes environment-specific tool-name blacklists derived from the AgentHarm training distribution rather than semantic harm patterns. The semantic layer (L3) alone shows no comparable gap—and the generic zero-shot LLM, which operates purely on semantic content without any symbolic rules, exhibits only a 3.4 pp cross-benchmark gap. This has concrete implications for defense design: symbolic rules provide near-perfect coverage within their training distribution but catastrophically fail to generalize to novel tool vocabularies. Upgrading from tool-name blacklists to parameter-semantic analysis (e.g., detecting attacker-controlled IBANs or exfiltration-destination emails regardless of the calling tool name) is the single most impactful architectural change for cross-environment generalization.

The AgentDojo structural blind spots (Section 4.2) reveal an orthogonal gap: malicious-parameter attacks succeed because the gate context does not include the user's task goal. The fundamental information needed to compare "user requested X, agent doing Y" is absent from runtime gate evaluation.

The SSDG experiments (Section 4.6) provide a more precise formulation: information deprivation is the structural factor, not semantic difficulty. P2 confirms that stripping context from an otherwise capable semantic defense reproduces symbolic-level gaps ($R = 3.60$). P1's rejection is equally informative—it shows that goal awareness is necessary but not sufficient: the defense must be architecturally capable of comparing goal to action, not merely receive the goal as a text fragment.

6.2 Symbolic vs. Semantic Generalization

The controlled experiment, together with the SSDG theoretical framework (Hypotheses 1–2), exposes a fundamental asymmetry between the two primary detection mechanisms in compositional safety architectures:

Symbolic rules (L1 Datalog) exhibit a bimodal generalization profile: near-perfect recall within the training distribution (100% on AgentHarm), and near-zero recall on out-of-distribution tool vocab-

ularies (3.7% on AgentDojo). This is structurally expected—Datalog rules encode extensional facts ("tool t is prohibited") rather than intensional harm semantics. Any tool name not in the rule set is trivially allowed.

Semantic reasoning (L3 gate and generic LLM) exhibits a flat generalization profile: below-perfect recall within distribution (62.7% for the generic LLM on AgentHarm), but stable recall across environments (59.3% on AgentDojo, gap = 3.4 pp). Semantic reasoning transfers because harm intent is encoded in goal descriptions—which remain semantically consistent across tool-vocabulary changes—rather than in tool names.

This asymmetry has a direct design implication for compositional architectures: the value of combining symbolic and semantic layers is not simply "two layers are better than one." The layers have *complementary generalization curves*: symbolic rules provide high-precision coverage in-distribution, and semantic reasoning maintains consistent coverage out-of-distribution. A system relying solely on symbolic rules will exhibit catastrophic degradation on novel tool environments; a system relying solely on semantic reasoning will accept a consistent recall ceiling but generalize robustly. The composable architecture's value is precisely that it avoids both failure modes.

6.3 Layer Complementarity on Hijacking

The Hijacking result (43.3% gate $\rightarrow$ 93.3% combined) provides direct evidence of architectural complementarity. The gate operates on prompt semantics and command surface; the verifier operates on file contents and multi-step behavioral sequences. These detection dimensions are orthogonal for prompt-injection attacks:

- When the injection is semantically explicit (e.g., "ignore previous instructions"), the gate catches it; the verifier sees no artifact yet.
- When the injection is embedded in a file artifact (e.g., a calendar event description or email body that the agent reads and acts upon), the semantic gate evaluates the *agent's action*, which may be semantically plausible; the verifier detects the injection fingerprint in the artifact content.
- When both signals are present, both layers fire (25% of cases).

This architectural insight—that prompt injection requires both *semantic* and *behavioral-sequence* detection—is a transferable design principle for agent safety systems.

6.4 Implications for Benchmark Design

The AgentDojo 3.7%$\rightarrow$14.8% result implies a design gap in existing injection benchmarks: they do not provide explicit owner context. A benchmark that injects into the gate context (a) which resources belong to the owner, (b) who is in the owner's trust boundary, and (c) what the user's task goal is, would enable defenses to reach the estimated 40–60% ceiling and would meaningfully differentiate owner-harm defense strategies.

We recommend future benchmarks include an *owner context manifest*: a structured specification of $R_{\mathcal{O}}$, $\mathcal{B}_{\mathcal{O}}$, and $\text{Auth}_{\mathcal{O}}$ provided to both the agent and any runtime safety system.

6.5 Limitations

1. **AgentDojo ceiling**: Without user task goal context, the semantic gate cannot compare "user requested X" to "agent doing Y", limiting achievable safety to $\sim$15% on malicious-parameter tasks.
2. **Benchmark independence**: The Owner-Harm Benchmark is author-constructed after system optimization. It is diagnostic, not validation evidence.
3. **Single annotator**: No inter-annotator agreement is reported for the Owner-Harm Benchmark labels.
4. **Structural boundary**: 4/60 Hijacking cases (SQL file injection, direct SSH key injection) are out of reach of current V-rules.

5. **Evaluation mode**: AgentDojo evaluation uses ground-truth mode without an adversarial attack LLM; real-world adaptive adversaries may achieve different results.
6. **Infrastructure Exposure FPR**: Rule OH-R3a over-triggers on benign external-channel configuration queries (20% FPR for C2), indicating the rule requires further refinement.
7. **SSDG pilot scale**: P1 and P2 experiments use $N = 12$ AgentDojo tasks and $N = 15$ AgentHarm scenarios. Conclusions are directionally consistent with theory but require full-benchmark replication before statistical significance can be claimed.

6.6 Future Work

Structured goal-action alignment. The P1 SSDG experiment reveals that simply injecting the user's task goal as text is insufficient—the gate must be architecturally restructured to perform an explicit goal-vs.-action comparison. A dedicated "goal alignment prompt" that contrasts the user's authorized intent against the agent's proposed action is the concrete next step, estimated to raise AgentDojo safety from 14.8% to 40–60%.

UEBA for AI agents. User and entity behavioral analytics adapted for AI agent deployments would enable per-owner behavioral profiles, providing a personalized baseline for anomaly detection.

Adaptive immunity. Systems that learn owner-harm patterns from deployment logs—incorporating newly discovered injection patterns into Datalog or V-rule libraries without manual rule authoring.

SQL-aware file content analysis. Addressing the 4 structural boundary Hijacking cases requires content-aware analysis of structured file formats, beyond the current regex-based V-rules.

Owner context manifest standard. A community standard for specifying $R_{\mathcal{O}}$, $\mathcal{B}_{\mathcal{O}}$, and $\text{Auth}_{\mathcal{O}}$ in a machine-readable format, to enable principled owner-harm benchmarking across systems.

Per-decision proof-obligation synthesis. Ongoing work extends this compositional gate from offline-authored Datalog rules to *per-decision runtime synthesis* of Datalog proof obligations by the LLM itself, admitted through a compile-time gate that rejects perturbation-invariant or decisive-primitive-less rules. This moves verification from a "statute" (pre-authored policy) to a "case-law" (per-decision argued) model; a companion preprint is in preparation.

7 Related Work

Agent safety and harm taxonomies. AgentHarm Andriushchenko et al. [2025] provides the most comprehensive generic harm benchmark for LLM agents, covering 11 categories from cybercrime to fraud. ToolEmu Ruan et al. [2024] evaluates LM agent risks in an LM-emulated sandbox with 7 risk types, partially overlapping our taxonomy but without an owner-centric perspective. AGrail Liu et al. [2025] proposes a hierarchical safety framework for agentic systems with adaptive safety detection. Our work differs by (a) formally defining the owner as the harm victim, and (b) providing quantitative evidence of the resulting defense gap.

Prompt injection. Greshake et al. Greshake et al. [2023] first systematically analyzed indirect prompt injection, demonstrating that LLM-integrated applications can be compromised through injected third-party content. AgentDojo Debenedetti et al. [2024] provides standardized evaluation of injection defenses across realistic agentic tasks. CaMeL Debenedetti et al. [2025] addresses prompt injection via capability-aware information flow, preventing data exfiltration at the language model level rather than the runtime layer.

Runtime safety systems. NeMo Guardrails Rebedea et al. [2023] provides programmable rails for LLM applications. Llama Guard Inan et al. [2023] implements LLM-based input/output safeguards. ShieldGemma Zeng et al. [2024] offers generative content moderation. R2Guard Chen et al. [2024] provides runtime guardrails specifically designed for agent tool calls, and AgentMonitor Yu et al. [2024] offers real-time behavior monitoring. BIPIA Yi et al. [2023] benchmarks indirect prompt injection attacks complementary to our analysis. These systems operate primarily on content classification; none model owner-context dimensions ($R_{\mathcal{O}}$, $\mathcal{B}_{\mathcal{O}}$, $\text{Auth}_{\mathcal{O}}$). Guardrails AI Guardrails AI [2024] provides composable input/output guards but without owner-harm semantics. Doshi et al. Doshi et al. [2026] propose verifiable tool safety via type systems, complementary to our runtime detection approach.

Policy-driven agent enforcement (2025–2026). A rapidly growing body of work explores policy-driven runtime enforcement for LLM agents. PCAS Palumbo et al. [2026] compiles a Datalog-derived declarative policy language offline and enforces it through a reference monitor. AgentSpec Wang et al. [2026] provides a lightweight DSL of triggers, predicates, and enforcement actions. Pro2Guard He et al. [2025] adds probabilistic reachability via learned DTMCs over execution traces. Solver-Aided verification Roy et al. [2026] translates natural-language tool-use policies into SMT-LIB constraints offline and verifies each call with Z3. ShieldAgent Chen et al. [2025] constructs probabilistic rule circuits from policy documents with formal-verification tool support. GuardAgent Xiang et al. [2024] synthesizes a guardrail plan and executes it as code, and TrustAgent Hua et al. [2024] inserts an Agent Constitution post-planning check. These systems focus on policy expressiveness and offline authorability; our contribution is orthogonal: the owner-context dimensions ($R_{\mathcal{O}}$, $\mathcal{B}_{\mathcal{O}}$, $\text{Auth}_{\mathcal{O}}$) must be *inputs* to any such policy regardless of its authoring method, yet are absent from the input vocabularies of all systems surveyed here.

Exfiltration and covert channels. Rehberger Rehberger [2024] demonstrated ASCII smuggling via Microsoft 365 Copilot, encoding sensitive data in Markdown rendered by the victim's browser. The PromptArmor Slack AI disclosure PromptArmor [2024] demonstrated credential exfiltration via injected prompts. Abdelnabi et al. Abdelnabi et al. [2024] provide a systematic analysis of adversarial attacks on LLM-based agents.

Post-gate auditing. Intrusion detection systems Liao et al. [2013] provide a conceptual precedent for behavioral sequence auditing after an initial gate, applied here to AI agent tool-call sequences.

Provably safe systems. Tegmark and Omohundro Tegmark and Omohundro [2023] argue that only formally verifiable systems can achieve controllable AGI, motivating the Datalog layer's emphasis on decidable rule-based reasoning.

8 Conclusion

Owner-harm is a systematically under-studied threat category in AI agent safety. We formalize it with eight categories and precise definitions, grounded in real incidents from 2023–2026. We demonstrate the defense gap quantitatively: a compositional safety system achieving 100% on generic criminal harm (AgentHarm) scores only 3.7%/14.8% on prompt-injection-mediated owner harm (AgentDojo), with specific structural causes attributed to each failure mode. We show that compositional detection—combining a semantic gate with a deterministic post-audit verifier—achieves 93.3% on the hardest sub-category (Hijacking), with the two layers operating on orthogonal detection signals.

The core finding is conceptually simple: existing defenses are owner-context-blind. An action is harmful to an owner not because of its content in isolation, but because of how it relates to the owner's resources, trust boundary, and authorization scope. Until safety systems model these context dimensions explicitly, owner-harm will remain a systematic blind spot—even as generic harm detection reaches saturation.

Author Contributions

D.Z. conceived the Owner-Harm threat model, designed and implemented the four-layer Nous system, ran all experiments on AgentHarm and AgentDojo, and drafted the manuscript. Y.J. (knowledge graph specialist, Tongji University) contributed to the initial research design discussions, provided critical review of the symbolic–semantic architecture and the owner-harm category taxonomy, and reviewed final manuscript revisions. Both authors approved the final version and are accountable for the integrity of the work.

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